

Advances in AI-Based Toxicity Prediction for Drug Discovery

Targeted Research Synthesis for survey paper

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Abstract

Drug toxicity is a major reason why promising drug candidates fail during preclinical studies, clinical trials, or after market approval. Artificial intelligence (AI) can support early toxicity screening by learning patterns from molecular structures, biological experiments, and known toxic compounds. This research synthesis examines recent developments in AI-based drug toxicity prediction using studies published between 2023 and 2026. Sixteen methods are compared for liver toxicity, heart toxicity, neurotoxicity, mutagenicity, genotoxicity, and carcinogenicity. Although recent machine learning and deep learning models report strong performance, their real-world reliability is limited by inconsistent datasets, class imbalance, weak external validation, limited biological context, and poor interpretability. This paper proposes an explainable multimodal and multi-task framework that combines molecular graphs, physicochemical properties, metabolites, off-target interactions, and biological-response data. The proposed framework would also provide uncertainty estimates and warnings when a prediction may be unreliable.

Index Terms

artificial intelligence, machine learning, drug discovery, toxicity prediction, graph neural network, explainable AI, drug safety

I. PROBLEM DEFINITION

The main bottleneck is that many AI toxicity models perform well on their original datasets but become less reliable for new and chemically different drug candidates. Toxicity datasets are often small, imbalanced, inconsistent, and collected using different experimental procedures. Many models also depend mainly on molecular structure and do not include dose, metabolism, exposure time, organ condition,

or human biological response. Random dataset splitting may further exaggerate performance by placing closely related molecules in both the training and testing groups [1]–[4].

Clinical impact: An incorrect prediction can directly affect patient safety. A false-negative result may allow a harmful compound to enter clinical trials, while a false-positive result may remove a potentially useful treatment. More dependable early prediction of liver, heart, kidney, brain, genetic, and cancer-related toxicity could reduce harmful adverse effects, unnecessary animal testing, late-stage drug failure, development costs, and delays in delivering safe medicines to patients [1]–[3].

II. STATE-OF-THE-ART ANALYSIS

Recent methods are moving from traditional machine learning, such as random forest, support vector machine, logistic regression, and gradient boosting, toward graph neural networks, attention mechanisms, active learning, and multimodal feature fusion. Graph-based models are useful because molecules can naturally be represented as atoms connected by chemical bonds. Attention mechanisms can also highlight atoms or molecular regions that strongly influence a toxicity prediction [1], [2].

General platforms such as ADMETLab 3.0, ProTox 3.0, and Deep-PK predict several toxicity and pharmacokinetic properties, while specialized systems focus on liver injury, hERG-related heart toxicity, neurotoxicity, mutagenicity, genotoxicity, or carcinogenicity. The reported results should not be treated as a direct ranking because the studies used different datasets, endpoints, class distributions, and validation methods. External validation, uncertainty estimation, applicability-domain analysis, and understandable explanations are also important when judging model reliability [1]–[3].

TABLE I: Comparison of Twenty Recent Papers on AI-Based Toxicity Prediction

Paper/Method	Endpoint or Scope	Input and AI Approach	Reported Result	Main Strength	Main Limitation
Lee <i>et al.</i> [1]	Review of multiple toxicity endpoints	Reviews RF, SVM, XGBoost, neural networks, GNNs, transformers, descriptors, and molecular graphs	No single performance score	Provides a recent overview of datasets, models, endpoints, and evaluation methods	A review paper; it does not introduce or experimentally validate one new prediction model
Zhang <i>et al.</i> [2]	Review of acute, organ-specific, genetic, and cancer toxicity	Reviews machine learning, deep learning, transfer learning, multimodal fusion, databases, and toxicity tools	No single performance score	Provides a broad framework covering databases, endpoints, challenges, and future directions	A review paper; comparisons depend on results reported by other studies
Gianquinto <i>et al.</i> [3]	Integrative drug safety and predictive toxicology	Reviews QSAR, molecular modeling, machine learning, multi-omics, toxicokinetics, and advanced in vitro systems	No single performance score	Connects computational prediction with biological mechanisms and clinical translation	A review paper; no single model or common benchmark dataset is evaluated
Ali [4]	Review of AI-driven drug-toxicity prediction	Reviews QSAR, machine learning, deep learning, toxicity databases, algorithms, and software tools	No single performance score	Explains major AI approaches, challenges, and future opportunities	Provides limited original quantitative comparison or external model validation

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Paper/Method	Endpoint or Scope	Input and AI Approach	Reported Result	Main Strength	Main Limitation
ADMETLab 3.0 [5]	Broad ADMET and toxicity	Molecular graphs with directed message-passing neural networks	hERG AUROC $\approx$ 0.94	Predicts 119 endpoints and provides uncertainty estimates and structural alerts	Benchmark results do not confirm performance in real clinical settings
ProTox 3.0 [6]	Organ, clinical, and general toxicity	Machine learning, chemical similarity, molecular targets, initiating events, and adverse outcome pathways	Endpoint-dependent	Provides broad toxicity coverage and some mechanistic information	Dataset size, quality, and predictive performance differ across endpoints
Deep-PK [7]	Thirty-five toxicity and pharmacokinetic endpoints	Graph neural network using molecular structures and physicochemical descriptors	Endpoint-dependent	Highlights molecular subgraphs that influence individual predictions	Contains limited dose, patient, exposure, and clinical information
InterDILI [8]	Drug-induced liver injury	Fingerprints and descriptors with random forest, LightGBM, logistic regression, and neural networks	AUROC 0.97; AUPRC 0.95; accuracy 0.90	Combines strong prediction performance with feature and attention explanations	Performance decreased when a different data-splitting method was applied
DILI Predictor [9]	Drug-induced liver injury	Fingerprints, descriptors, predicted pharmacokinetic properties, proxy labels, and random forest	AUROC 0.63	Combines predicted and experimental information in an accessible tool	Reported discrimination is moderate and limited to a specific liver endpoint
GeoDILI [10]	Drug-induced liver injury	Three-dimensional molecular graphs with a geometry-based graph neural network	AUROC 0.908; accuracy 0.975	Provides atom-level explanations of possible toxic molecular regions	Requires stronger independent, temporal, and prospective validation
OvA-QSTR [11]	Drug-induced liver injury	PaDEL descriptors with Bayesian networks, decision trees, and random forests	AUPRC 0.718–0.869	Uses relatively simple and understandable machine-learning models	Hand-crafted descriptors may fail to capture complex biological mechanisms
Rao <i>et al.</i> [12]	Severity of drug-induced liver injury	Physicochemical descriptors and off-target profiles with RF, SVM, neural networks, and logistic regression	AUROC 0.88; specificity 0.90	Predicts toxicity severity and includes off-target interaction information	Depends on the availability and accuracy of curated off-target data
hERGBoost [13]	hERG inhibition and heart toxicity	Molecular descriptors and fingerprints with XGBoost classification and regression	Accuracy 0.814; MCC 0.614; R^2 0.622	Predicts both toxicity class and continuous IC50 values	Provides limited explanation of why a molecule is predicted as risky
hERGAT [14]	hERG inhibition	Molecular graphs, fingerprints, descriptors, graph attention networks, and gated recurrent units	AUROC 0.907; AUPRC 0.904	Provides atom-level and molecule-level attention explanations	hERG inhibition represents only one possible mechanism of cardiotoxicity
AttenhERG [15]	hERG inhibition	Molecular graph representation with an AttentiveFP graph neural network	AUROC 0.835; MCC 0.543	Provides prediction uncertainty and atom-level attention information	Reports lower discrimination than several other recent hERG models
DMFGAM [16]	hERG inhibition	Extended-connectivity and atom-pair fingerprints combined with graph attention networks	AUROC 0.894; accuracy 0.817	Combines molecular-graph and fingerprint-based information	Does not include clinical, metabolic, dose, or omics information
NeuTox 2.0 [17]	Blood-brain barrier effects, neuronal damage, and neurotoxicity	Molecular graphs, fingerprints, descriptors, DMPNN, MFBERT, and multimodal feature fusion	AUROC up to 0.971	Predicts several related neurotoxicity outcomes using feature fusion	Most input features are still derived mainly from molecular structure
muTOX-AL [18]	Ames mutagenicity	Extended-connectivity fingerprints, MACCS keys, descriptors, CNN, and active learning	AUROC 0.909	Achieves strong performance while using a smaller amount of labelled training data	Requires repeated expert or laboratory annotation during active learning
Fournier <i>et al.</i> [19]	Genotoxicity across several experimental assays	Two-dimensional descriptors and MACCS keys with similarity search and extremely randomized trees	AUROC approximately 0.921–0.972	Covers several genotoxicity assays rather than only one test	No complete executable prediction system was reported in the reviewed material
DCAMCP [20]	Carcinogenicity	Multiple molecular fingerprints with capsule networks and self-attention routing	AUROC 0.793; accuracy 0.718	Uses attention mechanisms and reports balanced classification measurements	Performance is moderate and available carcinogenicity data remain limited

Important note: The first four rows are review papers that summarize many models and research trends. The remaining sixteen rows represent individual prediction platforms or model-development studies. Therefore, review papers do not report one common accuracy value. AUROC and AUPRC values closer to 1 indicate stronger discrimination. Accuracy represents the proportion of correct predictions, while MCC ranges from -1 to 1 . Results from different datasets, endpoints, and validation procedures are not directly comparable.

III. TECHNICAL GAPS

The first major gap is data quality and generalization. Public toxicity datasets may contain duplicate compounds, missing values, conflicting labels, and highly imbalanced classes. Random train-test splitting can place structurally similar compounds in both groups, making model performance appear stronger than it is. Many studies still lack scaffold-based, time-based, external, and prospective validation using newly tested compounds [1]–[3].

The second gap is limited biological context. Most models mainly use molecular structure, although real toxicity also depends on dose, exposure duration, metabolism, age, genetics, organ condition, disease status, and interactions with other drugs. A compound may appear safe in its original form but produce harmful metabolites after being processed by the body [2], [3].

Another limitation is interpretability. Deep-learning models may behave as black boxes and make confident predictions outside the chemical area represented by their training data. Attention maps and feature-importance methods provide useful

clues, but they do not always prove that the highlighted feature is the true biological cause of toxicity. These weaknesses reduce trust among researchers, clinicians, toxicologists, and regulatory authorities [1], [2], [4].

IV. PROPOSED DIRECTION

A useful improvement would be an explainable multimodal and multi-task toxicity-prediction framework. A graph neural network would learn the molecular structure, while separate model branches would process physicochemical descriptors, predicted metabolites, off-target interactions, dose information, and available transcriptomic or cellular-response data. Combining these data types could provide a more complete description of how a drug behaves inside the body [1]–[3].

The model would predict several endpoints together, including liver injury, heart toxicity, neurotoxicity, kidney toxicity, mutagenicity, and carcinogenicity. Multi-task learning would allow endpoints with larger datasets to support related endpoints with fewer examples. The system should also report uncertainty and provide an applicability-domain warning

whenever a new molecule is very different from the compounds used during training.

Evaluation should use scaffold-based, temporal, and independent external test sets. Newly generated laboratory results could return to the model through active learning. Compounds with uncertain or conflicting predictions would receive priority for experimental testing. This human-in-the-loop process would allow AI to support toxicology experiments rather than attempting to replace them.

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